

Harry Nguyen

+1 470-667-9000 | nguyenharry2662@gmail.com | linkedin.com/in/harrynguyen26 | github.com/HarryNguyen2662

SUMMARY

Backend engineer and new grad (May 2027) building distributed production systems at TiMoto AI and Google Chrome. Ships reliable APIs and infrastructure at scale. Open to relocation to New York City.

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Primary engineer for backend, cloud infrastructure, and ML serving on a 3-person team – designed, deployed, and operate **distributed production systems** end-to-end
- Designed and maintained **gRPC inter-service layer**; resolved production deadlock under concurrent calls by tracing shared resource acquisition conflicts and redesigning call sequencing – **100% evaluation success rate** at **sub-50ms p99**
- Architected **vLLM inference engine with PagedAttention** for LLM serving – eliminated KV cache memory fragmentation under concurrent load; deployed with continuous batching achieving **zero OOM failures** at production traffic
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Maintained distributed production systems across gRPC, PostgreSQL, and ML serving layers – triaged and resolved incidents via root-cause analysis, documented runbooks, and applied post-mortem learnings; participated in on-call rotation

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protocol Buffers over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern decoupling UI state propagation – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Collaborated with Chrome infrastructure team on design documents and code reviews; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor